Population Data Entry Guide

How to enter ex-situ population data so it surfaces on the Coordinator Dashboard at `/dashboard/coordinator`.

Admin URL: https://api.malagasyfishes.org/admin/

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0. What data lives where

At the top level, five models connect to make a coordinator view:

Model	What it is	Who writes it
`Species`	The ~79 endemics	Seeded from CSV; curated by admin
`Institution`	A zoo / aquarium / hobbyist keeper / research org	You, through admin
`ExSituPopulation`	"Institution X holds species Y" — one per species-institution pair	You, through admin
`HoldingRecord`	Optional census snapshot for an `ExSituPopulation`, with counts on a date	You, or auto when you update population counts
`CoordinatedProgram`	Formal program (AZA SSP / EAZA EEP / CARES / independent) around a species	You, through admin
`Transfer`	Movement of animals between institutions — lifecycle from proposed → completed	You, through admin

`CoordinatedProgram` and `Transfer` land in Gate 4. This guide covers both the Gate 3 models (Species / Institution / ExSituPopulation / HoldingRecord) and the Gate 4 additions.

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1. Where each panel gets its data

Panel	Reads from	Driven by which field(s)
1 — Coverage gap	`Species` ← *absence of* `ExSituPopulation`	`Species.iucn_status ∈ {CR, EN, VU}` AND no `ExSituPopulation` rows reference the species. Endemic toggle filters `Species.endemic_status`.
2 — Studbook status	`ExSituPopulation` grouped by species	`studbook_managed` (boolean) + `breeding_status` (enum). See §5 "How a species lands in each bucket."
3 — Sex-ratio risk	`ExSituPopulation` (one row per population)	`count_male`, `count_female`, `count_unsexed`. Flag thresholds in code: skew > 1:4, or unsexed fraction > 50%.
4 — Stale census	`ExSituPopulation` + `HoldingRecord`	`max(last_census_date, most recent HoldingRecord.date)`. Flag if that signal is > 12 months old, or never set.

DD companion card on Panel 1 reads `Species.iucn_status = DD` — no population data needed.

Species records are already seeded. You don't need to add species; the ~79 endemics from `madagascar_freshwater_fish_seed.csv` are already in the registry with IUCN / CARES / endemic status populated. Dashboard panels light up as soon as you start adding `Institution` and `ExSituPopulation` rows.

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2. Prerequisites

- A Tier 3+ staff account (or superuser) in Django admin.
- A shared understanding of what counts as an "institution" — see §4 for the CARES-as-individuals pattern.

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3. Institution types

`Institution.institution_type` is an enum. Pick the closest match — this doesn't affect any gating today, but it does determine the label the panels and directory show next to the name.

Value	Intended for
`zoo`	Accredited zoo
`aquarium`	Accredited public aquarium
`research_org`	University lab, research station
`hobbyist_program`	Coordinated hobbyist program (the CARES umbrella, Citizen Conservation, regional breeding networks)
`hobbyist_keeper`	**Individual hobbyist** — a single person maintaining a population. Use this for CARES participants you're tracking as individuals.
`ngo`	Conservation NGO running captive work
`government`	Government fisheries agency

Individual CARES keepers should go in as `hobbyist_keeper`, not `hobbyist_program` — see §4.

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4. CARES individual keepers (the common case)

The pattern: each CARES hobbyist keeper is their own `Institution` row with `institution_type = hobbyist_keeper`. Each population that keeper holds is its own `ExSituPopulation` row linking species → keeper.

This keeps per-keeper fragility visible on the dashboard — Panel 2 "Holdings only" and Panel 3 "Sex-ratio risk" show the keeper name and flag them independently, which is the whole point of treating them as individuals rather than a bucket.

Minimum fields for a keeper

Field	What to enter

`name`	How the keeper wants to be credited. Full name, handle, or program ID — your call, but be consistent.
`institution_type`	`hobbyist_keeper`
`country`	Country of the keeper
`city`	Optional; omit if privacy-sensitive
`contact_email`	Visible **Tier 3+ only** per the model's help_text. Safe to fill in if the keeper consented. Leave blank otherwise.
`website`	Usually blank for individuals
`zims_member` / `eaza_member` / `aza_member`	Leave unchecked for individual keepers
`species360_id`	Leave blank

Suggested name convention

For individual keepers, a legible format beats a perfect one. Some options that read well in dashboard tables:

- `J. Smith (CARES)` — initial + surname + program tag
- `Smith Household – CARES` — household attribution
- `CARES #4172` — program ID, if the keeper prefers anonymity

Pick a convention and stick to it per data-entry session. The name is what appears on the dashboard's sex-ratio and stale-census tables, so readability for a coordinator scanning at a workshop matters more than formal correctness.

Privacy posture

- `Institution.contact_email` is only serialized at Tier 3+ — public site hides it. Fine to store keeper emails here **with consent**.
- Keeper *addresses* are not in the model. If you need to track location, use `country` + `city` only.
- If a keeper wants to be fully anonymous, use a program-ID name and leave `contact_email` blank. The dashboard only ever displays `Institution.name`.

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5. Workflow: record an ex-situ population

For every species-holder pair you want to surface on the dashboard:

1. Confirm the **species** exists in **Species → Species**. It almost certainly does; the registry has every described and undescribed endemic.
2. Confirm the **institution/keeper** exists in **Populations → Institutions**. If not, add it per §4.
3. Go to **Populations → Ex situ populations → Add**.
4. Fill in:

Field	Notes
`species`	Autocomplete — type scientific name

`institution`	Autocomplete — type institution/keeper name
`count_total`	Total count; required-ish (nullable in the schema but Panel 3 reads it)
`count_male` / `count_female` / `count_unsexed`	M.F.U — fill all three when known. Dashboard shows `male.female.unsexed`. Leave all three blank for unsexed-only or unknown populations.
`date_established`	When this population was founded at this institution. Optional but useful.
`founding_source`	e.g. "wild-caught Nosivolo 2019", "split from Cologne F2". Free text.
`breeding_status`	`breeding` / `non-breeding` / `unknown`. See §6 for how this interacts with `studbook_managed`.
`studbook_managed`	Check **only** if there's a formal studbook keeper assigned and a coordinated plan. For CARES populations this is usually **unchecked** — that's what puts them in Panel 2's "breeding, not studbook" bucket, which is intentional.
`last_census_date`	Date you last counted the population. **Panel 4 watches this field** — if it's more than 12 months old, the population shows up as stale.
`notes`	Free text. Anything a coordinator should know.

5. Save.
6. (Optional but recommended) On the species or institution list view, run the **"Revalidate public pages"** action. Without this, the public site's ISR cache may take up to an hour to show the new population. Admin saves on Species/Genus revalidate automatically; saves on Institution/ExSituPopulation do not yet — this is a minor gap to be closed later.

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6. How a species lands in each Panel 2 bucket

For any species with ≥1 `ExSituPopulation` row:

If...	Bucket
At least one population has `studbook_managed = True`	**Studbook-managed**
None studbook, but ≥1 has `breeding_status = 'breeding'`	**Breeding, not studbook**
None studbook, none breeding	**Holdings only**

Species with zero `ExSituPopulation` rows are counted as **No captive population** — those are the same species Panel 1 lists.

If a CARES keeper has a breeding population of a CR species and isn't part of a studbook, that population lands the species in "Breeding, not studbook" — which is the signal the four-bucket panel exists to surface. Don't check `studbook_managed` to "promote" a keeper into the tidier bucket; leave it accurate.

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7. Workflow: logging a census snapshot

You have two options for recording "I counted this population on date X and got these numbers":

Option A — just update the ExSituPopulation (simplest)

Edit the `ExSituPopulation` row directly: change `count_total` / `count_male` / `count_female` / `count_unsexed`, update `last_census_date` to today, save. Panel 4 will no longer flag it as stale.

Use this when: you don't need a history — you only care about the current state.

Option B — add a HoldingRecord (for time-series)

Open the `ExSituPopulation` row → scroll to the **Holding records** inline at the bottom → add a row with the census date and new counts → save.

`HoldingRecord` is ordered by `-date`, so the most recent one is what Panel 4 treats as the "latest holding" signal. You can leave `last_census_date` stale as long as there's a recent `HoldingRecord`; Panel 4 uses `max(last_census_date, latest holding)`.

Use this when: you want to track how a population has changed over time (mortality events, breeding events, new acquisitions).

For the workshop demo, Option A is sufficient. Option B becomes more useful once you're collecting CARES data quarterly and want to see population trajectories.

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8. Troubleshooting

A population I added isn't showing up on the dashboard

1. Admin save succeeded? Check that the row exists at **Populations → Ex situ populations**.
2. The Coordinator Dashboard page is force-dynamic (re-fetches every request), but the Django API response itself may be cached. Wait a minute and refresh.
3. If you're looking at Panel 2 (studbook status), confirm the species you'd expect shows up in a bucket by looking at the four counts at the top — a new holding increments the bucket count even if the species list under it is long.

A species I added a population for is still in Panel 1

Panel 1 shows species with **zero** populations. If you added a population and the species still appears, reload the page (force refresh). If it persists, double-check the `ExSituPopulation` row actually points at the right species (autocomplete sometimes selects a near-match).

Panel 4 says a population is stale but I just updated it

Ensure you updated `last_census_date`, not just `count_total`. Panel 4 ignores count changes — it watches the date field.

Panel 3 doesn't flag a population I expect to see

Confirm `count_male`, `count_female`, and `count_unsexed` are all populated (not null). If all three are null, Panel 3 can't compute a ratio and skips the row.

8.5 Gate 4 additions — CoordinatedProgram + Transfer

These two models capture the "who runs this" and "what's moving" layers above `ExSituPopulation`. Added in Gate 4 Phase 1. Admin surfaces live alongside the existing population models.

CoordinatedProgram

One row per species per program framework. A species can appear in multiple programs simultaneously (e.g. AZA SSP and CARES), but only once per `program_type`.

When to add one: when a species has a formal breeding / conservation program tracking it (AZA SSP, EAZA EEP, CARES priority listing, or an independent regional program).

Admin: `/admin/populations/coordinatedprogram/add/`

Key fields:

- `species` — autocomplete
- `program_type` — `ssp` / `eep` / `cares` / `independent` / `other`
- `name` — human-readable, shown on the dashboard. Format guidance: *"AZA SSP: Common Name"*, *"EAZA EEP: Common Name"*, *"CARES: Common Name"*
- `status` — `planning` / `active` / `paused` / `deprecated`
- `coordinating_institution` — who holds the studbook
- `studbook_keeper` — if a named user is assigned
- `enrolled_institutions` (M2M) — partner zoos / keepers
- `target_population_size`, `plan_summary`, `plan_document_url`, `start_date`, `next_review_date` — plan-level metadata

Transfer

One row per planned or completed animal movement between two institutions.

When to add one: any time animals move between `Institution` rows — whether that's a formal SSP transfer, a hobbyist moving a founder pair, or an accession from the wild (with the collecting institution as source). Use `status` to indicate where the transfer is in its lifecycle.

Admin: `/admin/populations/transfer/add/`

Key fields:

- `species` / `source_institution` / `destination_institution` — all autocomplete. Source ≠ destination (enforced at the DB level).
- `status` — `proposed` / `approved` / `in_transit` / `completed` / `cancelled`
- `proposed_date` — when the transfer was first logged. Required.
- `planned_date` — when it's *scheduled* to happen. Optional.
- `actual_date` — when it *actually* completed. Set this when status moves to `completed`.
- `count_male` / `count_female` / `count_unsexed` — M.F.U convention
- `cites_reference` — permit number for CITES-listed species. Blank otherwise.
- `coordinated_program` — link to the program this transfer serves, if applicable.
- `notes` — anything else. Holding pens, quarantine details, etc.

`created_by` auto-fills to the admin user on save — not editable.

Why these are separate from `ExSituPopulation`

`ExSituPopulation` is *current state* (counts per species per institution today). `Transfer` is *movement* — even if the counts don't change (e.g. a 1:1 swap), the transfer still happened. Both matter to a coordinator. Don't manually edit `ExSituPopulation` counts to reflect a transfer; log the `Transfer` row and update the population's counts separately when the animals arrive.

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9. Bulk import — `seed_populations`

For bulk CARES loads, use the `seed_populations` management command instead of clicking each row through admin.

CSV shape

One row = one population. Institutions are deduplicated by `institution_name` — three rows with the same name create one Institution and three populations attached to it.

Column	Required	Notes
`institution_name`	yes	The dedup key
`institution_type`	no	Default `hobbyist_keeper`. Enum: `zoo`, `aquarium`, `research_org`, `hobbyist_program`, `hobbyist_keeper`, `ngo`, `government`
`country`	no	Default `"Unknown"`
`city`	no	
`contact_email`	no	Tier 3+ only on display
`species_scientific_name`	yes	Must already exist in the registry
`count_total` / `count_male` / `count_female` / `count_unsexed`	no	Integers, blanks OK
`breeding_status`	no	Enum: `breeding` / `non-breeding` / `unknown`. Default `unknown`
`studbook_managed`	no	Boolean. Accepts `true/false/yes/no/1/0`. Default `false`
`last_census_date` / `date_established`	no	`YYYY-MM-DD`
`founding_source`, `notes`	no	Free text

Convention for CARES hobbyist rows: `institution_type=hobbyist_keeper`, `studbook_managed=false`, set `last_census_date` to the date your CARES data was collected.

Example row

```
institution_name,institution_type,country,species_scientific_name,count_total,count_ma
J. Smith (CARES),hobbyist_keeper,United States,Paretroplus
menarambo,10,4,5,1,breeding,false,2026-04-15
```

Running it on staging

SSH in and drop the CSV at `data/seed/populations.csv` on the host (SCP from your laptop or edit with `nano` on the server):

```
ssh deploy@46.224.196.197
cd /home/deploy/madagascarfish/deploy/staging

# Dry-run first — parse + validate, roll back all DB changes
docker compose exec -T web python manage.py seed_populations \
  --csv /data/seed/populations.csv --dry-run

# Review the output, then the real run
docker compose exec -T web python manage.py seed_populations \
  --csv /data/seed/populations.csv
```

The command is **idempotent**: re-running with the same CSV updates existing rows rather than duplicating them. Dedup keys are `Institution.name` and `(species, institution)` for populations.

Error handling

- Rows with invalid enums or missing required fields are **skipped individually** — the rest of the CSV still loads. Errors print to stderr (first 50, then a suppressed-count line).
- Rows referencing a species scientific name that's not in the registry are skipped with a `species not found` error.

What it does NOT do

- Doesn't touch `HoldingRecord` — populations get current-state counts, historical census snapshots are admin-only.
- Doesn't touch `CoordinatedProgram` or `Transfer` — those stay admin-only.

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10. See also

- `OPERATIONS.md` — staging runbook, shared secrets, revalidate webhook
- `ADMIN GUIDE.md` — general admin reference, not population-specific
- `CLAUDE.md` — access tiers, sensitive data rules